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ELECTRONIC CIGARETTE PACKAGE AND ELECTRONIC CIGARETTE E-LIQUID FILLING PROCESS

Abstract

A box assembly includes a body portion; a tray disposed within the body portion; and a cover disposed over the tray and body portion. The body portion includes a first filling port at one end, and the tray includes a second filling port aligned with the first filling port. A rack assembly can hold a plurality of box assemblies. The rack assembly can be held by an e-liquid filling apparatus for filling a device contained within each box assembly with e-liquid.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure generally relates to atomization and vaporizing devices, and more particularly, to e-liquid filling of an electronic cigarette device.

BACKGROUND

[0002] Electronic cigarette (also known as “E-cigarette”), or vaping, devices can be used to deliver nicotine, cannabis (THC, CBD), flavorings, chemicals, and other substances. These devices are known by many different names and come in many shapes, sizes and device types. These devices may also be referred to as “E-cigs”, “Vapes”, “Vape pens,” “dab pens,” “dab rigs.” “Tanks,” “Mods,” “Pod-Mods,” and the like. Use of e-cigarette, or vaping, products is sometimes referred to as “vaping.”

[0003] Typically, a conventional e-cigarette device or electronic cigarette device includes several basic components: a cartridge (also referred to as a reservoir or pod), an atomizer (or atomizer core) including a heating element, a power source (e.g., a battery), and a mouthpiece. The cartridge (or reservoir or pod) can hold various substances. The cartridge may be pre-loaded with these substances, and sold with or separate from the rest of the e-cigarette device. One particular substance is a liquid solution (sometimes referred to as “e-liquid” or “e-juice”). In one particular example, the liquid solution may contain varying amounts of one or more substances including, but not limited to, nicotine, botanical oil (e.g., hemp oil, cannabis oil (e.g., THC, CBD)), flavorings, and/or other chemicals. Some conventional e-cigarette devices may not use a cartridge to hold the liquid solution. Instead, these e-cigarette devices include a reservoir built-into the device for containing the liquid solution, and into which the liquid solution can be filled. In many e-cigarette devices, puffing by a user results in an airflow entering the electronic cigarette device. As the air flows into the electronic cigarette device, the generated airflow will trigger an airflow sensor, and thereby activate the heating element of the atomizer. The electric heating element, disposed within an atomization channel of the atomizer core, starts to heat the e-liquid, and generate aerosol or vapor, which then flows out through the atomization channel under the drive of the airflow, and the resulting aerosol or vapor travels to the mouthpiece where the aerosol or vapor is then inhaled by the user.

[0004] In order to prevent conventional electronic cigarettes from leaking e-liquid during long-distance transportation or to prevent e-liquid in electronic cigarettes from deterioration, etc. (e.g., due to causes such as climate differences (e.g., temperature, humidity, air pressure, etc.) in different regions), some conventional electronic cigarettes are delivered directly to dealers or distributors after being assembled and produced by manufacturers, and then transferred to a point of sale for e-liquid filling (i.e., these conventional electronic cigarettes are not filled with e-liquid until after they are delivered to a point of sale (e.g., a smoke shop, a convenience store, a supermarket, an online retailer, etc.). In this method of e-liquid filling, the conventional electronic cigarettes need to be removed from their packaging (e.g., boxes and the like), placed into an e-liquid filling device for e-liquid filling, and then placed back into their packaging after being filled with e-liquid. The e-liquid filling process can be cumbersome and inefficient due to electronic cigarettes needing to be removed from their packaging, the packaging placed in a location where the packaging will not be damaged while the electronic cigarettes are filled with e-liquid, and then placing the electronic cigarettes back into their packaging (without damaging the packaging in the process).

[0005] Accordingly, there is a need for an improved process for filling electronic cigarettes with e-liquid. There is a further need for an improved process for filling electronic cigarettes with e-liquid which combines packaging and e-liquid filling. There is an additional need for an improved process for filling electronic cigarettes with e-liquid such that the electronic cigarettes are packaged prior to being filled with e-liquid. There is also a need for packaging for electronic cigarettes that allows the electronic cigarettes to be filled with e-liquid while the electronic cigarettes are in their packaging. There is yet a further need for eliminating the need to remove electronic cigarettes from their packaging prior to the electronic cigarettes being filled with e-liquid and then re-package electronic cigarettes after being filled with e-liquid. There is an additional need for an improved process for filling electronic cigarettes with e-liquid that improves production efficiency. There is a further need for an improved e-liquid filling device or fixture for filling electronic cigarettes with e-

liquid that is easier to manufacture, assemble, disassemble, adjust, and maintain. The present invention satisfies these needs and provides other related advantages.

SUMMARY OF THE INVENTION

[0006] The present invention provides an improved process for filling electronic cigarettes with e-liquid. The present invention provides an improved process for filling electronic cigarettes with e-liquid which combines packaging and e-liquid filling. The present invention provides an improved process for filling electronic cigarettes with e-liquid such that the electronic cigarettes are packaged prior to being filled with e-liquid. The present invention provides packaging for electronic cigarettes that allows the electronic cigarettes to be filled with e-liquid while the electronic cigarettes are in their packaging. The present invention eliminates the need to remove electronic cigarettes from their packaging prior to the electronic cigarettes being filled with e-liquid and re-package electronic cigarettes after being filled with e-liquid. The present invention provides an improved process for filling electronic cigarettes with e-liquid that improves production efficiency. The present invention provides an improved e-liquid filling device or fixture for filling electronic cigarettes with e-liquid that is easier to manufacture, assemble, adjust, and maintain. The present invention satisfies these needs and provides other related advantages, including that no further packaging is required after e-liquid filling. Furthermore, scratches and collisions on an outer surface of the electronic cigarette devices during e-liquid filling are also avoided, such that not only the production efficiency is improved, but also the quality of products is ensured. Improved production efficiency is achieved by optimizing the process.

[0007] In accordance with an embodiment of the present invention, the electronic cigarette e-liquid filling process includes the steps of preparing a self-sealing electronic cigarette and a perforated packaging box; placing the self-sealing electronic cigarette into the perforated packaging box; placing the packaged plurality of self-sealing electronic cigarettes into limit holes of a rack assembly; and mounting the rack assembly to a position for e-liquid filling in an e-liquid filling apparatus for e-liquid filling.

[0008] In accordance with an embodiment of the present invention, a box assembly includes a body portion, a tray portion disposed within the body portion, and a cover portion disposed over the tray portion and body portion. The body portion includes a first filling port at one end, and the tray portion includes a second filling port aligned with the first filling port.

[0009] In accordance with another embodiment of the present invention, the second filling port tray includes an open end of the tray portion aligned with the first filling port.

[0010] In accordance with an additional embodiment of the present invention, the body portion includes an accommodating cavity sized and shaped to match a size and shape of the tray portion.

[0011] In accordance with a further embodiment of the present invention, the cover portion is configured to removably engage the body portion.

[0012] In accordance with a still further embodiment of the present invention, the tray portion includes an interior portion sized and shaped to receive a device. The device includes a filling port aligned with the second filling port of the tray portion when disposed within the tray portion.

[0013] In accordance with still another embodiment of the present invention, the filling port of the device is aligned with the first filling port of the body portion.

[0014] In accordance with yet another embodiment of the present invention, the cover portion includes a plurality of recessed perimeter walls set back on all sides from respective side edges of the cover portion. The perimeter walls extend downward from a bottom surface of the cover portion and are positioned to be received within a recessed portion of the body portion such that a portion of a bottom surface of the cover portion abuts against a top surface of the body portion.

[0015] In accordance with an embodiment of the present invention, the cover includes a plurality of bulges configured to press-fit engage an interior portion of the body portion.

[0016] In accordance with another embodiment of the present invention, a rack assembly is configured for holding at least one box assembly device in position for providing a pre-determined

amount of e-liquid to a device having a filling port contained within the at least one box assembly. The rack assembly includes an upper cover, a lower cover, and a limit member disposed between the upper and lower covers. The upper cover, lower cover, and limit member are configured to vertically align the at least one box assembly. At least one of the upper and lower covers includes at least one filling hole. The at least one box assembly includes a body portion, a tray portion disposed within the body portion, and a cover portion disposed over the tray portion and body portion. The body portion includes a first filling port at one end, the tray portion includes a second filling port, and the first and second filling ports are aligned with the at least one filling hole and the filling port of the device.

[0017] In accordance with an additional embodiment of the present invention, the rack assembly further includes first and second support members. The first and second support members engage the upper and lower covers.

[0018] In accordance with a further embodiment of the present invention, the upper cover includes at least one upper limit groove, with the at least one upper limit groove configured to receive and engage a portion of the at least one box assembly at a first end of the at least one box assembly.

[0019] In accordance with a still further embodiment of the present invention, the lower cover includes at least one lower limit groove, with the at least one lower limit groove configured to receive and engage a portion of the at least one box assembly at a second end of the at least one box assembly.

[0020] In accordance with still another embodiment of the present invention, a method for providing e-liquid to at least one device having a filling port contained within a box assembly. The at least one device includes a reservoir in communication with the filling port. The at least one box assembly includes a body portion, a tray portion disposed within the body portion, and a cover portion disposed over the tray portion and body portion. The body portion includes a first filling port at one end, the tray portion includes a second filling port, and the first and second filling ports are aligned with the at least one filling hole and the filling port of the device. The method includes providing an e-liquid filling apparatus that includes a workbench, an e-liquid filling assembly configured to provide e-liquid to the reservoir of the at least one device contained within the box assembly, a moving assembly configured to adjust a filling portion of the e-liquid filling assembly and the box assembly relative to each other, and a control assembly configured to control operation of the e-liquid filling apparatus. The method also includes configuring the control assembly to fill the at least one device contained within the box assembly with e-liquid, including setting a pre-determined amount of e-liquid the e-liquid filling assembly provides to the at least one device. The method additionally includes positioning the at least one device in a rack assembly configured to hold the at least one device. The method further includes operationally engaging the rack assembly with the workbench. The method also includes filling the at least one device contained within the box assembly with the pre-determined amount of e-liquid.

[0021] This brief summary has been provided so that the nature of the invention may be understood quickly. Additional aspects and advantages of the present invention will be given in part in the following more detailed description, taken in conjunction with the accompanying drawings, which can become apparent from the following description, which illustrate, by way of example, the principles of the invention or be understood through practice of the present invention. Any drawings contained herein constitute a part of this specification and include exemplary embodiments of the present invention and illustrate various objects and features thereof.

Description

DESCRIPTION OF THE FIGURES

[0022] The various present embodiments now will be discussed in detail with an emphasis on

highlighting the advantageous features with reference to the drawings of various embodiments. The illustrated embodiments are intended to illustrate, but not to limit the invention. These drawings include the following figures, in which like numerals indicate like parts: The above and/or additional aspects and advantages of the present invention will be apparent and easily understood from the descriptions of the embodiments with reference to the following drawings, wherein:

- [0023] FIG. 1 illustrates a front left, top perspective view of a box assembly according to an embodiment of the present invention;
- [0024] FIG. 2A illustrates an exploded view of the box assembly of FIG. 1;
- [0025] FIG. 2B illustrates a perspective view of a tray of the box assembly of FIG. 2A;
- [0026] FIG. 2C illustrates a perspective view of an alternate tray for use with the box assembly of FIG. 2A;
- [0027] FIG. 2D illustrates a perspective view of a box body portion and an underside of a cover portion of the box assembly of FIG. 2A that illustrates a fastening mechanism for removably engaging the cover with the box body;
- [0028] FIG. 3 illustrates a front, right top perspective view of an electronic cigarette for use with the box assembly of FIG. 2A;
- [0029] FIG. 4 illustrates a cross-sectional view of the electronic cigarette of FIG. 3 (arrows indicate the direction of air flow through the electronic cigarette during vaping);
- [0030] FIG. 5 illustrates a front, right top perspective view of another electronic cigarette for use with the box assembly of FIG. 2A;
- [0031] FIG. 6A illustrates a front, right top perspective view of an additional electronic cigarette for use with the box assembly of FIG. 2A;
- [0032] FIG. 6B illustrates the electronic cigarette of FIG. 6A positioned within the tray of FIG. 2C;
- [0033] FIG. 7 illustrates a front, right top perspective view of yet another electronic cigarette for use with the box assembly of FIG. 2A;
- [0034] FIG. 8 illustrates a perspective view of an electronic cigarette disposed within the box assembly of FIG. 1 being filled with e-liquid by an e-liquid filling head (with the cover of the box assembly removed to illustrate a needle of the e-liquid filling head disposed within a filling port of the electronic cigarette);
- [0035] FIG. 9 illustrates a top perspective view of a rack assembly configured to hold a plurality of box assemblies, with each box assembly configured to hold an electronic cigarette or atomizer;
- [0036] FIG. 10 illustrates a bottom perspective view of the rack assembly of FIG. 9 with the e-liquid filling head disposed below a leftmost filling hole on a bottom cover of the rack assembly (an arrow indicates the direction the e-liquid filling head moves from left to right during a process of filling electronic cigarettes (not shown) held by the rack assembly);
- [0037] FIG. 11 illustrates an front elevation view of the rack assembly and e-liquid filling head of FIG. 10 (an arrow indicates the direction the e-liquid filling head moves from left to right during a process of filling electronic cigarettes (not shown) held by the rack assembly);
- [0038] FIG. 12 illustrates an embodiment of an e-liquid filling apparatus holding the rack assembly of FIG. 11 in position during filling of the electronic cigarettes within the box assemblies held within the rack assembly; and
- [0039] FIG. 13 illustrates an embodiment of a process of placing electronic cigarettes within box assemblies, placing box assemblies in a rack assembly, and positioning the rack assembly in an e-liquid filling apparatus configured to fill the electronic cigarettes held within the box assemblies held by the rack assembly with e-liquid.

DETAILED DESCRIPTION

[0040] The following description is provided to enable any person skilled in the art to make and use the invention and sets forth the best modes contemplated by the inventor of carrying out their invention. Various modifications, however, will remain readily apparent to those skilled in the art, since the general principles of the present invention have been defined herein specifically to

provide an electronic cigarette assembly. The following detailed description describes the present embodiments, with reference to the accompanying drawings. In the drawings, reference numbers label elements of the present embodiments. These reference numbers are reproduced below in connection with the discussion of the corresponding drawing features. It is to be understood that the figures and descriptions of the present invention have been simplified to illustrate elements that are relevant for a clear understanding of the present invention, while eliminating, for the purpose of clarity, many other elements found in, without limitation, electronic cigarette assemblies, electronic cigarette packaging, and apparatus for filling electronic cigarette assemblies with e-liquid. Those of ordinary skill in the pertinent arts may recognize that other elements and/or steps are desirable and/or required in implementing the present invention. However, because such elements and steps are well known in the art, and because they do not facilitate a better understanding of the present invention, a discussion of such elements and steps is not provided herein. The disclosure herein is directed to all such variations and modifications to such elements and methods known to those skilled in the pertinent arts.

[0041] Embodiments of the present invention are described in detail hereinafter, and illustrations of the embodiments are shown in the drawings, wherein identical or similar reference numerals denote identical or similar elements or elements having the same or similar functions. The embodiments described hereinafter with reference to the drawings are exemplary and only intended to explain the present invention, and cannot be understood as limiting the present invention.

[0042] With reference to FIGS. **1-13**, embodiments of the present invention provide for improved packaging or box assemblies for holding electronic cigarette devices or atomizers during various stages including, without limitation, transport, filling with e-liquid, and sale. The instant application incorporates by reference herein the contents of U.S. patent application Ser. No. 18/418,099, entitled “E-Liquid Filling Device”, filed Jan. 19, 2024.

[0043] As seen in FIGS. **1** and **2A**, an embodiment of a perforated packaging box or box assembly **100** includes a box body or body portion **110**, a tray or tray portion **120** disposed within the body portion **110**, and a cover or cover portion **130** disposed over the tray **120** and body portion **110**. The body portion **110** includes a first e-liquid filling port or first filling port **111** located in a sidewall **112** at one end of the body portion **110**. As seen in FIGS. **2A** and **2B**, the tray **120** is provided with a second e-liquid filling port or second filling port **121** located in a sidewall **122** at one end of the tray portion **120**. The second e-liquid filling port **121** is aligned with, and adjacent to, the first e-liquid filling port **111**. As seen in FIG. **2A**, the filling ports **111**, **121** may be the same shape and/or size. However, the filling ports **111**, **121** may be different shapes and/or sizes. As seen in FIG. **2A**, the filling ports **111**, **121** are roughly racetrack/oval shaped but the filling ports **111**, **121** may be any desired shape/size including, without limitation, circular, triangular, rectangular, square, or any polygonal shape.

[0044] Alternatively, an embodiment of the tray **120** may be configured with an open side, as seen in FIG. **2C** adjacent to and aligned with the side of the body portion **110** where the first e-liquid filling port **111** is located (i.e., allowing e-liquid filling to be achieved with the open end of the tray portion **120** acting as the second e-liquid filling port **121**).

[0045] The cover portion **130** is configured to removably engage the body portion **110**. The body portion **110**, the tray portion **120**, and the cover portion **130** may be made from various materials including, without limitation, plastic, metal, cardboard, wood, Carbon-containing material, or the like. The body portion **110**, the tray portion **120**, and the cover portion **130** may be made from the same material, or the body portion **110** and the cover portion **130** may be made from a harder material than the tray portion **120**. For example, the body portion **110** and the cover portion **130** may both be made from Acrylic (or Polymethyl Methacrylate (PMMA)), and the tray portion **120** may be made from Polypropylene (PP). The body portion **110**, the tray portion **120**, and the cover portion **130** may be transparent, translucent, or opaque, as desired. For example, in one embodiment, the tray portion **120** may be transparent while the body portion **110** and the cover

portion **130** are opaque. In another embodiment, however, the tray portion **120** could be opaque while the body portion **110** and the cover portion **130** are transparent.

[0046] The tray portion **120** is formed with a groove or accommodating cavity A, and the structure/shape of the groove or accommodating cavity A is sized/shaped to match the size and shape of an electronic device, such as an electronic cigarette or an atomizer. The body portion **110** is formed with a groove or accommodating cavity B, and the structure/shape of the groove or accommodating cavity B is sized/shaped to match the size and shape of the tray portion **120**, with the tray portion **120** contained within the body portion **110**. The groove or accommodating cavity A of the tray portion **120** comprises an interior portion sized and shaped to receive the electronic device **200, 300, 400, 500**. The filling ports **111, 121** are positioned to align with each other and with a filling port **201, 301, 401, 501** of the electronic cigarette device **200, 300, 400, 500** such that an e-liquid filling needle or the like may pass through the filling ports **111, 121** and into the filling port **201, 301, 401, 501** of the electronic cigarette device **200, 300, 400, 500**. The location of the filling ports **111, 121** may vary depending on the location of the filling port **201, 301, 401, 501** of the electronic cigarette device **200, 300, 400, 500** to be placed within the box assembly **100**. For example, the aligned filling ports **111, 121** may be positioned on any side of the body portion **110** and tray portion **120** of the box assembly **100** necessary such that the filling ports **111, 121** are aligned with the filling port **201, 301, 401, 501** of the electronic cigarette device **200, 300, 400, 500**. Alternatively, in the event that the filling port **201, 301, 401, 501** of the electronic cigarette device **200, 300, 400, 500** is located on a side of the electronic cigarette device **200, 300, 400, 500** directly under the cover portion **130**, the cover portion **130** could include a filling port aligned with the filling port of the electronic cigarette device (without the need for filling ports in the body portion **110** and tray portion **120**).

[0047] The cover portion **130** is removably engaged to the box body by a fastening mechanism **131** that provides for engagement of the cover portion **130** with the body portion **110** when the cover portion **130** is positioned over the body portion **110**. The cover portion **130** can be separated from the body portion **110** by removing the cover portion **130** from engagement (e.g., press-fit engagement) with the body portion **110**. The body portion **110** also includes at least one slot or hole **114** on the sidewall **112** of the body portion **110** for a user to engage with a fingernail or tool to use as leverage when disengaging the cover portion **130** from the body portion **110**. In one embodiment of the fastening mechanism **131**, the fastening mechanism **131** includes a plurality of bulges **132** formed on at least two (2) of the four (4) recessed perimeter sidewalls **133** of the cover portion **130** such that when the cover portion **130** is disposed over and engaged with the body portion **110**, the bulges **132** tightly engage respective sidewalls of the body portion **110**. The number and location of the bulges **132** on each sidewall **133** can be as desired (e.g., at least one (1) bulge **132** may be formed on each sidewall **133** of the cover portion **130**). In this embodiment, the body portion **110** includes a recessed portion **113** disposed along an upper part of the interior portion of the body portion **110**. The recessed perimeter walls **133** are set back on all sides from respective side edges **134** of the cover portion **130**, and extends downward from a bottom surface **136** of the cover portion **130**. The recessed perimeter walls **133** are positioned to be received within the recessed portion **113** of the body portion **110** such that an outer portion **135** of a bottom surface **136** of the cover portion **130** (disposed between the perimeter walls **133** and the side edges **134** of the cover portion **130**) abuts against a top surface **115** of the sidewalls of the body portion **110**. The depth of the recessed portion **113** from the top surface **115** of the sidewalls (e.g., sidewall **112**) is greater than or about equal to the distance the perimeter walls **133** extend downwardly from the bottom surface **136** of the cover portion **130**. The height of the sidewalls (e.g., sidewall **122**) of the tray portion **120** is less than or about equal to the distance between a bottom surface **116** of the groove B and the recessed portion **113** so that the top surfaces of the sidewalls (e.g., sidewall **122**) of the tray portion **120** do not abut against bottom surfaces of the perimeter walls **133** of the cover portion **130** in a manner that prevents engagement of the cover portion **130** and the body portion **110**. In

one embodiment, two (2) bulges **132** are illustrated on one side of the indented perimeter wall **133** of the cover portion **130** in FIG. 2D with two (2) bulges **132** identically positioned on the indented perimeter wall **133** on an opposite end of the cover portion **130** (in this embodiment, the body portion **110** does not include the at least one slot or hole **114** on a sidewall **112** of the body portion **110** corresponding to the location of the bulges **132**). When the cover portion **130** is covered over the body portion, the bulges **132** tightly engage (i.e., press fit (or interference fit) engages) the nearest sidewall (e.g., sidewall **112**) of the body portion **110**. In the alternative, various other fastening mechanisms can be used including, without limitation, snap-fit engagement, clips, magnets embedded at both ends for connection, etc. In another alternative, the cover portion **130** and the body portion **110** are fixed to each other or otherwise permanently engaged with each other but the side of the body portion **110** with the filling port **111** can be opened/closed. For example, the side of the body portion **110** with the filling port **111** can be, without limitation, removably attached to the body portion **110**, hingedly attached to the body portion **110** so that the side can be opened/closed (securable in the closed position), or otherwise adjusted between open and closed positions in order to provide access to the interior of the body portion **110**.

[0048] The structure of the box assembly **100** allows an electronic cigarette held within to be filled with e-liquid while the electronic cigarette is disposed within the box assembly **100**. The electronic cigarettes may be filled with e-liquid manually, individually or in batches. The electronic cigarettes may also be filled with e-liquid by a e-liquid filling apparatus **700**, such as those disclosed in U.S. patent application Ser. No. 18/418,099, entitled “E-Liquid Filling Device”, filed Jan. 19, 2024, the contents of which are incorporated by reference herein in their entirety.

[0049] The box assembly **100** may be used with any electronic cigarette, including self-sealing electronic cigarettes, such as those described in more detail below. As seen in FIGS. 3-7, various designs of electronic cigarette devices may be used in conjunction with the box assembly **100**, and four (4) embodiments of self-sealing electronic cigarette devices **200**, **300**, **400**, **500** are described herein for purposes of illustration only. For purposes of illustration, and to avoid unnecessary repetition, only the structure of the electronic cigarette device **200** will be described in further detail, with the other electronic cigarettes **300**, **400**, **500** including the same/similar features. As seen in FIG. 4, the electronic cigarette device **200** includes a housing **211**, a mouthpiece portion **202**, a self-sealing elastomer element **210**, an atomizer core **230**, a seal seat **240**, a battery **250**, a sensor bracket **260**, an gas flow sensor **270**, a circuit board **280**, an atomization base **290**, and a USB interface **291**.

[0050] The mouthpiece portion **202** is arranged at one end of the housing **211**, and the mouthpiece portion **202** includes a smoking port or port **201** which provides an outlet for vapor to exit the electronic cigarette device **200** and access to an e-liquid storage cavity **220** disposed within the housing **211**. The mouthpiece portion **202** is generally disposed within the mouth of a user while the user is inhaling vaporized e-liquid. An end of the e-liquid storage cavity **220** facing the port **201** is sealed by the self-sealing elastomer element **210**. An end of the e-liquid storage cavity **220** facing away from the port **201** is sealed by the seal seat **240**. The self-sealing elastomer element **210** is provided with an e-liquid filling guide groove **212**. The e-liquid filling guide groove **212** is used to facilitate an e-liquid filling needle (the e-liquid filling needle may be part of a syringe for manual filling of e-liquid, or part of an e-liquid filling head **710** such as seen in FIGS. 8 and 10) piercing the self-sealing elastomer element **210** along the e-liquid filling guide groove **212** to fill e-liquid into the e-liquid storage cavity **220**. The e-liquid filling guide groove **212** is aligned with and corresponds to the port **201**. The port **201** is aligned with and corresponds to the first e-liquid filling port **111** and the second e-liquid filling port **121**. During e-liquid filling, the e-liquid filling needle (either part of a syringe for manual filling of e-liquid, or part of an e-liquid filling head **710** such as seen in FIGS. 8 and 10) passes through the first e-liquid filling port **111**, the second e-liquid filling port **121**, the port **201** and the e-liquid filling guide groove **212** sequentially, and then pierces the self-sealing elastomer element **210** to fill e-liquid into the e-liquid storage cavity **220**. After the e-

liquid filling is completed, the e-liquid filling needle is pulled out, and the pierced part of the self-sealing elastomer element **210** automatically recovers and the puncture created by the filling needle self-seals based on the elasticity of the self-sealing elastomer element **210**. The self-sealing elastomer element **210** is made from a material (including, without limitation, rubber, silicone, latex, plastic or the like) that deforms under an external force and is capable of quickly recovering to its original shape after the external force is removed.

[0051] The atomizer core **230** is arranged inside the housing **211**. The atomizer core **230** includes a vapor guide tube **231**, an e-liquid guide element (not shown for clarity) and a heating element (not shown for clarity). The vapor guide tube **231** is provided with an e-liquid inlet hole (not shown for clarity) and a gas flow channel **214** through which vaporized e-liquid passes towards the port **201**. The e-liquid in the e-liquid storage cavity **220** enters the e-liquid guide element through the e-liquid inlet hole. The e-liquid guide element transfers the e-liquid entering the atomizer core **230** to the heating element for heating and atomization. One end of the vapor guide tube **231** is mounted on the seal seat **240** and the other end thereof is disposed within a bore extending through the self-sealing elastomer element **210** that opens to the port **201**. After the e-liquid is heated, the vapor generated is discharged through the vapor guide tube **231** to the port **201** for the user to smoke. FIG. 5 illustrates another self-sealing electronic cigarette device **300** having a mouthpiece portion **202** and a port **301**. FIG. 6A illustrates a further self-sealing electronic cigarette device **400** having a mouthpiece portion **202** and a port **401**. FIG. 7 illustrates an additional self-sealing electronic cigarette device **500** having a mouthpiece portion **202** and a port **501**.

[0052] The present application is illustrated with the electronic cigarette device **200**, and it should be understood that fitting between the box assembly **100** and the electronic cigarette device **200** (or any electronic cigarette device **300**, **400**, **500**) can be achieved by adjusting the size of the tray portion **120** or the overall size according to the size of the self-sealing electronic cigarette device **200**. For example, FIG. 6B illustrates the electronic cigarette device **400** positioned within the tray portion **120** of FIG. 2C.

[0053] By adjusting the tray portion **120** of the box assembly **100** to a particular self-sealing electronic cigarette device **200**, **300**, **400**, **500**, the e-liquid filling can be achieved without removing the self-sealing electronic cigarette device **200**, **300**, **400**, **500** from the box assembly **100**, and the self-sealing electronic cigarette device **200**, **300**, **400**, **500** and box assembly **100** act as one piece to be filled with e-liquid, which simplifies the e-liquid filling process, improves production efficiency, and also avoids scratches and collisions on an outer surface of the electronic cigarette device **200**, **300**, **400**, **500** during e-liquid filling, thus ensuring the quality and condition of the product at the time of sale.

[0054] The operating principle of the self-sealing electronic cigarette device (for example, electronic cigarette device **200**) is described below in combination with the direction of a gas flow (seen in arrows in FIG. 4). The gas flow enters the self-sealing electronic cigarette **200** from the atomization base **290**, sequentially flows along a gap between the battery **250** and the housing **211**, the seal seat **240** and a gas flow channel **213** within a lower part of the atomizer core **230**, carries the atomized e-liquid in the gas flow channel **214** of the smoke guide tube **231**, and finally is inhaled at the port **201** by the user. When the gas flow passes through the gas flow sensor **270**, the gas flow sensor **270** senses a gas pressure difference to control the heating of the heating element within the atomizer core **230**.

[0055] As seen in FIGS. 9-11, an e-liquid filling fixture or rack assembly **600** is illustrated, such as those disclosed in U.S. patent application Ser. No. 18/418,099, entitled "E-Liquid Filling Device", filed Jan. 19, 2024, the contents of which are incorporated by reference herein in their entirety. The rack assembly **600** includes a rack upper cover **610**, a limit member **620**, a rack lower cover **630**, a first support column (or first support member) **641**, and a second support column (or second support member) **642** (the first and second support columns **641**, **642** may be identical in shape and size; as illustrated, the first and second support columns **641**, **642** are generally cylindrical in

shape). The rack upper cover **610** has a pair of upper support holes **613**, with each support hole **613** respectively sized and shaped to engage an upper portion of one of the first and second support columns **641**, **642**, and the rack lower cover **630** has a pair of lower support holes **633** with each support hole **633** respectively sized and shaped to engage a lower portion of one of the first and second support columns **641**, **642**. The rack upper cover **610** includes a plurality of first limit grooves (not shown), with each first limit groove sized and shaped to receive and engage a portion of a respective box assembly **100** at a first end of the respective box assembly **100** (e.g., for purposes of illustration, there are sixteen (16) box assemblies **100** seen in FIG. **11**, and therefore sixteen (16) first limit grooves would be present. Likewise, the rack lower cover **630** includes a plurality of second limit grooves **631**, with each second limit groove **631** sized and shaped to receive and engage a portion of a respective box assembly **100** (e.g., for purposes of illustration, there are sixteen (16) box assemblies **100** seen in FIG. **11**, and therefore sixteen (16) second limit grooves **631** are illustrated in FIG. **9**). The first and second limit grooves **611**, **631** may be sized and shaped to specifically match the respective shapes of the first and second ends of a particular design of box assembly. While the illustrated rack assembly **600** shows a plurality of identically shaped box assemblies **100** (e.g., sixteen (16) box assemblies **100** seen in FIG. **11**), the rack assembly **600** may be designed to hold any number of box assemblies **100**. While the rack assembly **600** is seen holding box assemblies **100** of identical shape and size, the rack assembly **600** may be designed to hold a mix of box assemblies **100** of different shapes and sizes. The limit member **620** is provided with a plurality of limit holes **621** sized and shaped to engage at least a portion of the box assembly **100** between the first and second ends of the box assembly **100** (again, for purposes of illustration, sixteen (16) limit holes **621** are illustrated in FIG. **11** in conjunction with the sixteen (16) box assemblies **100** seen in FIG. **11**). The limit holes **621** are configured to limit the movement of each box assembly **100** in a horizontal direction (with the dimensions of the limiting holes **621** also able to be adjusted to provide a press-fit between each box assembly **100** and each limiting hole **621** to reduce relative vertical movement therebetween). The fixture upper cover **610** and the fixture lower cover **630** are configured to limit the movement of each box assembly **100** in a vertical direction. The limit member **620** may engage each box assembly **100** at a portion of the box assembly **100** generally mid-section (e.g., extending both sides of the mid-point of the box assembly **100** between the first and second ends of the box assembly **100**). In the alternative, the limit member **620** may engage each box assembly **100** at a portion of the box assembly **100** anywhere between the first end and the mid-section of the box assembly **100**, or the limit member **620** may engage each box assembly **100** at a portion of the box assembly **100** anywhere between the second end and the mid-section of the box assembly **100**. Alternatively, more than one limit member **620** may be used (e.g., one limit member **620** between the first end and the mid-section of a box assembly **100** and another limit member **620** between the second end and the mid-section of the box assembly **100**). The first limit groove, the second limit groove **631**, and the limit holes **621** are used for fixing the box assembly **100**, to avoid shifting, shaking, etc. of the box assembly **100** during an e-liquid filling process. Each limit hole **621** is aligned with a particular first limit groove and a particular second limit groove **631** when the rack assembly **600** is assembled. Each support column **641**, **642** has an upper end inserted into one of the upper support holes **613** and a lower end inserted into one of the lower support holes **633**. The rack assembly **600** can be assembled in various orders. For example, the box assemblies **100** may be inserted into their respective second limit grooves **631**, with the first and second support columns **641**, **642** then inserted into their respective lower support holes **633**. A lower end of each support column **641**, **642** is disposed within a respective lower support hole **633**. The limit member **620** can then be lowered around the box assemblies **100** such that an upper end of each box assembly **100** passes through a particular one of the limit holes **621** with the box assembly **100** held within the limit hole **621**. Each first limit groove of the rack upper cover **610** can be aligned with a first end of each box assembly **100**, and the rack upper cover **610** lowered downward such that the first end of each box

assembly **100** enters a respective one of the first limit grooves, and an upper end of each support column **641**, **642** enters into a respective upper support hole **613**. The foregoing is merely illustrative and can be adjusted as desired. For example, the process of assembling the rack assembly **600** can start with each box assembly **100** being inserted into a respective one of the limit holes **621**, with the box assembly **100** now being aligned for subsequent engagement with the rack upper cover **610** and the rack lower cover **630**. The support columns **641**, **642** may be fixedly connected with the rack lower cover **630** and detachably connected with the rack upper cover **610**. Alternatively, the support columns **641**, **642** may be detachably connected with both the rack upper cover **610** and the rack lower cover **630**. The upper cover **610**, the lower cover **630** and the support columns **641**, **642** may be made from various materials including, without limitation, plastic (e.g., PMMA (acrylic), metal (e.g., stainless steel), wood, or the like. The limit member **620** may be made from various materials including, without limitation, plastic (e.g., Expanded Polyethylene (EPE), Polystyrene (PS), biodegradable foam), metal, wood, or the like.

[0056] When an e-liquid filling needle (either part of a syringe for manual filling of e-liquid, or part of an e-liquid filling head **710** such as seen in FIGS. **8** and **10**) fills the e-liquid upwards with the box assemblies **100** disposed above the e-liquid filling needle (as seen in FIGS. **10-12**, the rack lower cover **630** includes plurality of e-liquid filling holes **632** through which at least a portion of the e-liquid filling needle (either part of a syringe for manual filling of e-liquid, or part of an e-liquid filling head **710** such as seen in FIGS. **8** and **10**) passes to inject e-liquid into one or more box assemblies **100**. The number of e-liquid filling holes **632** depends on the number of box assemblies **100** the rack assembly **600** is designed to hold. An e-liquid filling hole **632** is located at a bottom of each second limit groove **631** and extends between the bottom of the rack lower cover **630** and the bottom of the second limit groove **631**. The location of the e-liquid filling hole **632** is designed such that the e-liquid filling hole **632** aligns with the filling ports **111**, **121** of the box assemblies **100** (and the port **201**, **301**, **401**, **501** of the electronic cigarette device **200**, **300**, **400**, **500** disposed within the box assemblies **100**) when the second (or bottom) end of the box assembly **100** is inserted into the second limit groove **631**. When the e-liquid filling needle (either part of a syringe for manual filling of e-liquid, or part of an e-liquid filling head **710** such as seen in FIGS. **8** and **10**) is oriented such that the e-liquid filling needle is designed to fill the e-liquid downwards with the box assemblies **100** disposed below the e-liquid filling needle (as seen in FIG. **8**), the e-liquid filling needle is located above the rack assembly **600**, and the e-liquid filling holes **632** are located on the upper cover **610** and correspond to the first limit grooves. Again, at least a portion of the e-liquid filling needle passes through the e-liquid filling hole **632** to inject e-liquid into one or more box assemblies **100**. An e-liquid filling hole **632** is located at a top of each first limit groove and extends between the top of the rack upper cover **610** and the top of the first limit groove. The location of the e-liquid filling hole **632** is designed such that the e-liquid filling hole **632** aligns with the filling ports **111**, **121** of the box assemblies **100** (and the port **201**, **301**, **401**, **501** of the electronic cigarette device **200**, **300**, **400**, **500** disposed within the box assemblies **100**) when the first (or top) end of the box assembly **100** is inserted into the first limit groove. No matter how the box assembly **100** itself is oriented, the e-liquid filling ports **632** are aligned with and arranged opposite the first e-liquid filling port **111** and the second e-liquid filling port **121**. Thus, it is to be understood that the e-liquid filling ports **631** may be provided on one side of the rack upper cover **610** or the rack lower cover **630**, that is, the e-liquid filling ports **631** may be provided at an upper end or at a lower end, and the e-liquid filling direction of the e-liquid filling needle (either part of a syringe for manual filling of e-liquid, or part of an e-liquid filling head **710** such as seen in FIGS. **8** and **10**) adjusted accordingly.

[0057] As seen in FIG. **12**, the rack assembly **600** may be placed in an e-liquid filling apparatus **700**, such as those disclosed in U.S. patent application Ser. No. 18/418,099, entitled “E-Liquid Filling Device”, filed Jan. 19, 2024, the contents of which are incorporated by reference herein in their entirety. The e-liquid filling apparatus **700** includes a housing that supports or contains the rest

of the e-liquid filling apparatus **700**. The e-liquid filling apparatus **700** further includes an e-liquid filling assembly that includes e-liquid filling head **710**, a moving assembly **720**, a control assembly (or controller) **750**, and a rack clamping assembly **730** (configured to hold the rack assembly **600**). The e-liquid filling apparatus **700** is connected through a power cord (not shown) to an external power supply (e.g., a 110 volt power supply such as an electrical socket in a wall or the like). The power cord may be separate from, and configured to engage, the e-liquid filling apparatus **700**, and the e-liquid filling apparatus **700** includes an electrical connector (not shown) for removably engaging the power cord (with the power cord having a plug for removably engaging the power supply). In the alternative, the power cord may be fixedly connected to the e-liquid filling apparatus **700**. The e-liquid filling apparatus **700** may include a battery as a back-up power supply (e.g., the battery may be a rechargeable battery charged when the e-liquid filling apparatus **700** is connected to the power supply). One or more foot pads **800** are provided at the bottom of the e-liquid filling apparatus **700**, and can be adjusted in height to maintain the balance of the e-liquid filling apparatus **700**.

[0058] The control assembly **750** includes a user interface or display **751** that may be positioned anywhere desired on the housing of the apparatus **700** for ergonomics and convenience of the user. The display **751** can be configured as an input device. The controller or control assembly **750** controls the direction and distance of movement of the moving assembly **720**. The controller or control assembly **750** also controls the temperature of each part of a heat generating system or heating system by inputting instructions via the display **751**. The display **751** may be visible to an operator or user of the e-liquid filling apparatus **700**. The control assembly **750** also includes various electronics including a micro-controller unit (MCU) (e.g., the MCU including a processor, at least one memory module, at least one communication interface, peripherals, etc.) configured to allow a user to operate the e-liquid filling apparatus **700** by the display **751** providing a user interface, such as graphical user interface (GUI), to a user for operation of the e-liquid filling apparatus **700** and indicators on the display **751** to inform a user about various conditions of the e-liquid filling apparatus **700**. As mentioned, the e-liquid filling apparatus **700** may have a processor and/or memory configured to effect one or more steps or providing instructions for one or more steps to be performed by the e-liquid filling apparatus **700**, and/or the processor and/or memory may be capable of storing one or more instructions. In supplement to or alternative to the display **751**, the control assembly **750** may also include buttons/switches and various indicators (e.g., colored light emitting diodes associated with particular indicia on the housing that indicate a particular condition (e.g., power on, error, or the like). For example, to prevent unauthorized use of the apparatus **700**, the control assembly **750** may include a pin entry assembly providing four buttons/switches for entry of a four-digit personal identification number (PIN) to enable/disable operation of the apparatus **700**. Alternatively, the PIN may be entered through the GUI of the display **751**. In another example, an indicator (e.g., a red LED) could be used for indicating a “disabled” or “locked” condition of the apparatus **100**, and another indicator (e.g., a green LED) for indicating an “enabled” or “unlocked” condition of the apparatus **700**. Likewise, the control assembly **750** may include an on/start or “power” button/switch to power on the apparatus **700**. Other indicators may be displayed on the GUI or on the control assembly **750** including an error condition, power status, or the like. The control assembly **750** is connected to at least one pump (not shown), temperature sensors (not shown), the display **751**, and various other electrical machinery (e.g., drive motors within a Y-axis movement module and a Z-axis movement module, at least one heater, etc.).

[0059] The housing includes a workbench assembly or workbench **740** configured for holding the rack assembly **600** in position for filling at least one box assembly **100** (containing an electronic cigarette device) with e-liquid. The workbench **740** includes the rack clamp assembly (or clamp assembly) **730** configured for clamping the rack assembly **600** in position on the workbench **740** in order to fill the at least one box assembly **100** (containing an electronic cigarette device) held by

the rack assembly **600** with e-liquid. The clamp assembly **730** includes a pair of guides **761**, a compression member **762**, and a pair of knobs **763**. As illustrated, the knobs **763** have star-shaped heads but the shape of the head can be any desired shape (e.g., round, square, knurled, or the like). The clamp assembly **730** may include a number of fixing seats (not shown for clarity) configured to hold the rack assembly **600** in position on the workbench **740**. For example, the fixing seats define a limit area for placement of the rack assembly **600** on the workbench **740**. The limit area defined by the fixing seats define the bounds for placement of the rack lower cover **630** such that the fixing seats are configured to engage three (3) sides of the rack lower cover **630**. When the rack assembly **600** is put on the workbench **740**, three (3) sides of the rack lower cover **630** come into contact with the fixing seats. The rack assembly **600** can be easily removed from the workbench **740**, or can be easily installed on the workbench **740**.

[0060] When it is necessary to fill at least one box assembly **100** (containing an electronic cigarette device) with e-liquid, the at least one box assembly **100** (containing an electronic cigarette device) is placed into the rack assembly **600**, and then the rack assembly **600** is placed in the limit area defined by the fixing seats. The workbench **740** includes a strip-shaped (e.g., generally rectangular) through-hole or hollowed-out area (not shown) in the portion **765** of the workbench **740** on which the rack assembly **600** is placed. When filling at least one box assembly **100** (containing an electronic cigarette device) with e-liquid, the e-liquid filling needle of the e-liquid filling head **710** is moved from the bottom up from the hollowed-out area **425** to fill the electronic cigarette device held within the box assembly **100** fixed within the rack assembly **600**. The workbench **740** is designed so that all or a portion of the rack assembly **600** can be replaced as needed in order to secure various different-shaped box assemblies **100** (each containing an electronic cigarette device) for filling with a desired e-liquid. As noted above, the rack assembly **600** can be used in a situation where the e-liquid filing head **710** fills the e-liquid upwards, and in a situation where the e-liquid filing head **710** fills the e-liquid downwards.

[0061] When the e-liquid filing head **710** fills the e-liquid upwards, the structure of the rack assembly **600** includes the rack upper cover **610**, the limit member **620**, the rack lower cover **630**, the first support column **641** and the second support column **642**. The first support column **641** and the second support column **642** is mounted (e.g., removably or fixedly) on the clamp lower cover **630**, the rack upper cover **610** is mounted (e.g., removably) on the upper end of the first support column **641** and the second support column **642**. The plurality of first limit grooves is provided on the rack upper cover **610**, and the plurality of second limit grooves **631** is provided on the rack lower cover **630** for positioning the ends of the box assemblies **100**. As previously discussed, the limit member **620** is provided with a plurality of limit holes **621** which are sized and shape to engage with the box assemblies **100**. The box assemblies **100** may be “inverted” in the limit holes **621** (as well as “inverted” in the first and second limit grooves **611**, **631**) if the e-liquid filling port of the electronic cigarette(s) (or atomizers) contained within the box assemblies **100** is normally oriented “up” since the e-liquid filling head **710** is coming from “below” the electronic cigarette(s) (or atomizer(s)) held within the box assemblies **100**. Again, the number of the first limit grooves, the second limit grooves **631** and the limit holes **621** are the same, and the first limit grooves, the second limit grooves **631**, and the limit holes **621** cooperate to “fix” the box assemblies **100** in position for filling the electronic cigarette devices (held within the box assemblies **100**) with e-liquid. Again, each of the second limit grooves **631** includes an e-liquid filling hole **632** for the needle of the e-liquid filling head **710** to pass through and into the e-liquid filling port **201**, **301**, **401**, **501** of the electronic cigarette(s) (or atomizer(s)) to fill the electronic cigarette(s) (or atomizer(s)) **200**, **300**, **400**, **500** with e-liquid. When the e-liquid filing head **710** fills the box assemblies **100** (each containing an electronic cigarette device) with e-liquid downwards, the main modification of the rack assembly **600** is: the e-liquid filing holes **632** are provided on the rack upper cover **610** corresponding to the bottom wall of the first limit grooves, and the box assemblies **100** (each containing an electronic cigarette (or atomizer) are mounted “upright” in the limit holes

621 of the limit member 620 if the e-liquid filling port 201, 301, 401, 501 of the electronic cigarette(s) (or atomizers) 200, 300, 400, 500 is normally oriented “up”. The guide bars 761 may be made from various materials including, but not limited to, metal (e.g., iron, steel, aluminum, or the like). The compression member 762 can be made from various material including, but not limited to, metal (e.g., iron, steel, aluminum, or the like). The star-shaped heads or knobs 763 can be made from various material including, but not limited to, plastic, metal (e.g., iron, steel, aluminum, or the like), or the like. The pair of guides 761 are fixed to the workbench 730. The guides 761 may be in various shapes including, without limitation, rods, bars, and the like (for purposes of illustration, the guides 761 are shown in the form of rods). The compression member 762 is sleeved onto the guides 761, with each guide 761 passing through a particular through-hole in the compression member 762. The compression member 762 is configured to be moved up or down along the guides 761 by turning the knobs 763 either clockwise or counterclockwise. When the rack assembly 600 is placed on the workbench 740, a user adapts the compression member 762 to the upper end of the rack assembly 600 (i.e., the rack upper cover 610) by moving the compression member 762 up or down using the knobs 763 so as to allow the rack assembly 600 to be positioned in the limit area of the workbench 740 by the user with a bottom surface of the compression member 762 being above a top surface of the rack upper cover 610, and the user then adjusting the compression member 762 downwards until the bottom surface of the compression member 762 contacts the top surface of the rack upper cover 610 to sufficiently hold the rack assembly 600 in position in the limit area. The knob 763 includes a bolt with a handle. The bolt portion of each knob 763 extends into a through-hole in a respective side of the compression member 762 that intercepts the through-hole in the compression member 762 through which a respective one of the guides 761 extends such that an end of the bolt portion of the knob 763 is configured to frictionally engage the exterior of the guide 761, and hold the compression member 764 in a “fixed” vertical position. When it is necessary to “fix” the compression member 762 in a certain vertical position along the guides 761 (generally, when the bottom surface of the compression member 762 contacts the top surface of the rack upper cover 610 to sufficiently hold the rack assembly 600 in position in the limit area), the knobs 763 are rotated to make their respective bolt portions firmly abut against a respective guide 761, and “fix” the compression member 762 in the desired position. Each fixing seat is provided with a threaded hole (not shown) and is fixed to the workbench 740. When replacement of one or more components of the clamp fixing member assembly 730 is needed to accommodate the shape of a particular box assembly 100, the knobs 763 are loosened such that the compression member 762 can be removed from engagement with the guides 761, and the one or more components of the clamp fixing member assembly 730 (including the compression member 762) can be replaced.

[0062] The e-liquid filling assembly also includes an e-liquid storage container (not shown), the e-liquid filling head (including a hollow needle or filing portion) 710, and an e-liquid delivery pipeline (not shown) providing fluidic communication between the e-liquid storage container and the e-liquid filling head 710 so that e-liquid can move from the e-liquid storage container to the e-liquid filling head 710. The e-liquid storage container may come in various forms including, without limitation, a tank-shaped container having an e-liquid filling hole (not shown) formed in an upper portion thereof with the e-liquid filling hole used for adding e-liquid into the e-liquid storage container to replenish the amount of e-liquid in the e-liquid storage container (also referred to as an e-liquid storage tank or e-liquid storage reservoir). The volume of the e-liquid storage container can be any desired amount. For example, the container may be sized to hold 2000 milliliters (or 2 liters) of e-liquid, with the container usually filled with 1500 milliliters (or 1.5 liters) of e-liquid. The e-liquid is poured into the storage container for different periods of time depending on the operator's physical strength, ranging from 1 minute to 5 minutes. A valve, lid, or the like is disposed over the e-liquid filling hole of the e-liquid storage container to provide a seal between the interior of the storage container and the outside environment that is gas-impermeable and liquid-impermeable. An

e-liquid (e.g., oil) pump is disposed between the storage container and the e-liquid filling head **710**. The pump is configured to pressurize the e-liquid, and the e-liquid is transported along the e-liquid delivery pipeline (along which the e-liquid passes through the pump) to the e-liquid filling head **710**. The pump can come in various forms selected to meet the function of the pump including, without limitation, a gear pump, a centrifugal pump, etc.). The e-liquid filling apparatus **700** further includes a heating system (not shown) for heating the e-liquid to reduce the concentration of the e-liquid so as to ensure smoothness during e-liquid filling. The heating system includes a first heating unit, a second heating unit, and a third heating unit. The first heating unit is used for heating the e-liquid storage container. The first heating unit is provided in the e-liquid storage container, and is in direct or indirect contact with the e-liquid. The first heating unit can come in various forms including, without limitation, a thermal resistance type heat source. The second heating unit is used for heating the e-liquid filling head **710**. Like the first heating unit, the second heating unit can also come in various forms including, without limitation, a thermal resistance type heat source. The second heating unit is provided in the e-liquid filling head **710**, and is in direct or indirect contact with the e-liquid. The third heating unit is used for heating the e-liquid delivery pipeline. Like the first and second heating units, the third heating unit can also come in various forms including, without limitation, a thermal resistance type heat source. In particular, the third heating unit can be in the form of a thermal resistance wire spirally wound around the e-liquid delivery pipeline. The temperature of the heating system (including at least one of the first heating unit, the second heating unit, and the third heating unit) can be adjusted by a controller of the control assembly **750** to ensure that the e-liquid is always maintained at a suitable concentration during the e-liquid filling in order to reduce and/or prevent the e-liquid delivery pipeline from being blocked due to solidification of the e-liquid therewithin. The first, second, and third heating units of the heating system are connected to, and controlled by, the control assembly **750**. Control of the heating system (including the first, second, and third heating units) can be accessed through the display **751**. The storage container, the e-liquid filling pump, and the e-liquid filling head **710** are provided with temperature sensors to control/monitor the temperature of the storage container, the e-liquid filling pump, and the e-liquid filling head **710**. For example, the display **751** can list the temperature of the e-liquid pouring into the storage container, the temperature of the e-liquid at the storage container, the temperature of the e-liquid at the filling pump, and the temperature of the e-liquid at the filling head **710**. For each of the foregoing, the display **751** can list the current temperature of the e-liquid, and the temperature that the e-liquid is supposed to be. The user can adjust the settings for what the temperature of the e-liquid is supposed to be in the container, at the pump, and at the filling head **710**.

[0063] The e-liquid filling head **710** includes a hollow needle or filling portion through which the e-liquid passes, and an e-liquid filling valve in which a valve core is provided for controlling on/off of the e-liquid filling head **21** (i.e., controlling e-liquid moving into the needle or filling portion). The moving assembly **720** is provided on the housing, and is configured to move the e-liquid filling head **710** so as to align the e-liquid filling head **710** with the filling ports **111**, **121** of the box assemblies **100** and the e-liquid filling hole **201**, **301**, **401**, **501** of the electronic cigarette (or atomizer) **200**, **300**, **400**, **500** to be filled with the e-liquid. The moving assembly includes a Y-axis movement module for controlling a transverse movement of the e-liquid filling head **710** (or the workbench **740** and clamp fixing member assembly **730**). The illustrated apparatus **700** does not require an X-axis movement module configured for controlling a longitudinal movement of the e-liquid filling head **710** but an alternative embodiment may include an X-axis movement module including structure configured to control movement of the workbench **740** in the X direction. In the alternative, the moving assembly **720** can be used for moving the clamp fixing member assembly **730** (and rack assembly **600**) instead of the e-liquid filling head **710**. The Y-axis movement module comprises a Y-axis track and a Y-axis driving motor, and the Z-axis movement module includes a Z-axis track and a Z-axis driving motor. The Z-axis movement module is configured to control

longitudinal (vertical) movement of the workbench **740** and the clamp fixing member assembly **730**.

[0064] It can be understood that the control assembly **750** controls the Y-axis, and Z-axis movement modules that, in turn, control a movement of the e-liquid filling head **710** toward the box assemblies **100**) and, alternatively, a movement of the workbench **740** and clamp fixing member assembly **730** (and rack assembly **600**) toward the e-liquid filling head **710** with the purpose of aligning the e-liquid filling head **710** with the filling ports **111**, **121** (and e-liquid filling ports **201**, **301**, **401**, **501** of the electronic cigarette device **200**, **300**, **400**, **500** to be filled with the e-liquid that are held within the box assemblies **100**. In the alternative, the control assembly **750** may be connected to sensors that can detect if there are less than a full rack assembly **600** of box assemblies **100** (e.g., less than sixteen (16) box assemblies **100** where the rack assembly **600** can hold a maximum of sixteen (16) box assemblies **100** for filling), and the control assembly **50** moves the e-liquid filling head **710** only to those locations in the rack assembly **600** where a box assembly **100** is detected.

[0065] As the control assembly **750** controls movement of the e-liquid filing head **710**, a user enters a preset number of box assemblies **100** for filling in the control assembly **750** (e.g., via user interface on the display **751**) (each electronic cigarette is filled with a preset amount of e-liquid, with the preset amount selected or entered via the control assembly **750**). The control assembly **750** will control the e-liquid filling head **710** to fill the preset number of box assemblies **100** (i.e., fill the electronic cigarettes **200**, **300**, **400**, **500** held within the box assemblies **100**). The e-liquid filling head **710** may be configured with sensors (operatively connected to the control assembly **750**) to determine if the e-liquid filling head **710** is engaging a filling hole **201**, **301**, **401**, **501** of an electronic cigarette **200**, **300**, **400**, **500**. The control assembly **750** is programmed with an initial (or start) position of the e-liquid filling head **710**, the moving direction and the moving distances of the e-liquid filling head **710** to the positions along the rack assembly **600** where each individual box assembly **100** would be located, so as to ensure that the e-liquid filling head **710** accurately engages the filling port **201**, **301**, **401**, **501** of each electronic cigarette **200**, **300**, **400**, **500** (through the filling ports **111**, **121** of the box assemblies **100**) during actual operation. The moving assembly **720** includes a Y-axis movement module provided on the housing of the apparatus **700**. The Y-axis movement module includes a Y-axis track and a Y-axis driving motor. An e-liquid filling head **710** is provided on a sliding block of the Y-axis track such that the e-liquid filling head **710** is configured to move back and forth along the Y-axis. That is, the longitudinal (vertical) up and down movement of the workbench **740** moves a box assembly **100** from top to bottom, and the e-liquid filling head **710** (longitudinally stationary) fills the box assemblies **100** with e-liquid from bottom to top.

[0066] The workbench **740** is provided directly above the e-liquid filling head **710**, and the hollowed-out area is provided on a top surface of a portion **765** of the workbench **740**. The hollowed-out area is used for avoiding blockage between an e-liquid filling port **201**, **301**, **401**, **501** of the electronic cigarette(s) (or atomizer(s)) **200**, **300**, **400**, **500** and the e-liquid filling head **710**. The moving assembly **720** also includes a Z-axis movement module on the back of the workbench **740**, and the Z-axis movement module controls, by means of a guide shaft, the relative distance between the box assemblies **100** to be filled with the e-liquid to perform a reciprocating movement in a Z-axis direction. The Z-axis movement module is used for moving the workbench **740** to and from a position where a needle of the e-liquid filling head **710** is inserted within an e-liquid filling port **201**, **301**, **401**, **501** of an electronic cigarette (or atomizer) **200**, **300**, **400**, **500** held within a box assembly **100** held by the rack assembly **600**. By means of the cooperation of the Y-axis movement module and the Z-axis movement module, the e-liquid filling head **710** can sequentially perform e-liquid filling of the electronic cigarette (or atomizer) **200**, **300**, **400**, **500** held within a box assembly **100** held by the rack assembly **600** that are to be filled with an e-liquid. Operation of the moving assembly **720** is controlled by the control assembly **750**. A servomotor of the moving

assembly **720** controls an actuating arm to drive the e-liquid filling needle **710** to move in the Y-axis direction, move back and forth along the Z-axis at each e-liquid filling port **631** and perform e-liquid filling at the highest point, and then return to an initial position.

[0067] In use, as seen in FIGS. **8-13**, a number of electronic cigarette devices (e.g., electronic cigarette device **200**) are selected and prepared to be packaged and a matching number of box assemblies **100** compatible with the electronic cigarette device are selected and prepared for receiving electronic cigarette devices. As seen in FIG. **13**, up to sixteen (16) electronic cigarette devices (and a matching number of box assemblies **100**) may be selected as that is the illustrated capacity that can be held by the limit member **620** of the rack assembly **600**. The number of box assemblies **100** that can be held by the limit member **620** is limited only by the size of the limit area of the e-liquid filling apparatus **700** which, in turn, limits the dimensions of the rack assembly **600** which, in turn, limits the dimensions of the limit member **620**, and thereby limits the number of box assemblies **100** that can be held by the limit member **620** due to the dimensions of the individual box assemblies **100**. Again, it is only for purposes of illustration that the limit member **620** is shown holding only sixteen (16) box assemblies **100**. The type of tray portion **120** (e.g., FIG. **2B** or FIG. **2C**) used in the box assembly **100** depends on the shape of the electronic cigarette device selected for packaging. As seen in FIG. **13**, the electronic cigarette device **200** is placed within the tray portion **120** of the box assembly **100**, and the tray portion **120** is placed within the body portion **110**. The cover **130** is then removably engaged with the body portion **110**. Each box assembly **100** containing a self-sealing electronic cigarette **200** is then placed in a limit hole **621** in the limit member **620** of the e-liquid filling fixture or rack assembly **600**. If the needle of the e-liquid filling head **710** is positioned below the lower cover **630**, and filling the e-liquid into the electronic cigarettes upwards, then the box assemblies **100** are “inverted” such that the filling ports **111**, **121** (and the filling ports of the electronic cigarette devices (e.g., filling port **201** of electronic cigarette device **200**)) are facing downward. For illustration purposes, the limit holes **621** are filled from left to right (as the e-liquid liquid filling head **710** moves from left to right, as indicated by the arrows seen in FIGS. **10-11**). The e-liquid filling apparatus **700** can be programmed so the e-liquid filling head **710** fills the electronic cigarettes in any desired order, and to even skip limit holes **621** if, for example, a user desires to place less than sixteen (16) electronic cigarettes by placing an electronic cigarette in every other limit hole **621**, leave at least two empty limit holes **621** between placements of electronic cigarettes, etc. Once the electronic cigarettes have been placed within the limit member **620**, a user then assembles the rack assembly **600** (as described above).

[0068] As seen in FIG. **13**, the rack assembly **600** is moved into position on the workbench **740** such that the rack lower cover **630** contacts the fixing seats, as described above, and the rack assembly **600** is disposed within the limit area with the filling holes **632** of the rack lower cover **630** disposed over the strip-shaped through-hole or hollowed-out area on the portion **765** of the workbench **740**, and the e-liquid filling head **710** (as seen in FIG. **12**). The vertical height of the compression member **762** is then adjusted downward along the guides **761** by loosening the knobs **763** to disengage the respective bolt portion of the knobs **763** from contact with respective guides **761**. The compression member **762** is moved downward until the bottom surface of the compression member **762** contacts the upper surface of the rack upper cover **610**. The compression member **762** is pressed against the rack upper cover **610** with sufficient force to keep the rack assembly **600** pressed against the portion **765** of the workbench **740**, and fixed in place in the limit area of the workbench **740** so as to prevent the rack assembly **600** from moving out of the limit area of the workbench **740**. The knobs **763** are tightened to engage the bolt portions of the knobs **763** with respective guides **762** in order to maintain engagement of the compression member **762** with the rack assembly **600** and hold the rack assembly **600** in position in the limit area, as seen in FIGS. **12** and **13**.

[0069] Once the rack assembly **600** is in position, the user can start operation of the e-liquid filling apparatus **700**. By accessing the control assembly **750** (e.g., via the graphical user interface (GUI)

on the display **751**), the user can commence the filling operation. As seen in FIGS. **10-13**, the e-liquid filing head **710** is designed to fill the e-liquid upwards. The box assemblies **100** are inverted (such that the filling ports **111, 121** face in a downward direction), and the e-liquid filing head **710** fills the e-liquid upwards into the box assemblies **100**. The control assembly **750** of the e-liquid filling apparatus **700** aligns the e-liquid filling head **710** with the filling ports **111, 121** of the left-most box assembly **100** held by the rack assembly **600**. The control assembly **750** of the e-liquid filling device **700** uses the servo motor to control an actuating arm **711** (part of the Y-axis movement module) to drive the e-liquid filling needle **710** to move in the Y-axis direction. In this manner, the Y-axis movement module drives the e-liquid filing head **710** to move along the length direction of the rack assembly **600** (e.g., from left to right, for purposes of illustration), so that the e-liquid filing needle on the e-liquid filing head **710** is aligned with the e-liquid filing holes **632** of the rack lower cover **630**. The Z-axis movement module drives the workbench **740** to move up and down along the Z-axis, and when the needle of the e-liquid filling head **710** is aligned with a box assembly **100** containing an electronic cigarette (or atomizer) (e.g., electronic cigarette **200**) to be filled, the workbench **740** moves up and down along the Z axis and then drives the rack assembly **600** up and down to realize the needle of the e-liquid filing head **710** being inserted into the box assembly **100** (and electronic cigarette (or atomizer) contained therein) to fill the electronic cigarette (or atomizer) (e.g., electronic cigarette **200**) with e-liquid. The Y-axis movement module sequentially moves the e-liquid filling needle **710** under each e-liquid filling port **632** along the Y-axis while the Z-axis movement module moves the e-liquid filling head **710** up and down to complete e-liquid filling of each box assembly **100** through the first e-liquid filling port **111**, the second e-liquid filling port **121** and the e-liquid filling port **632**. When all the electronic cigarettes have been filled with e-liquid (each electronic cigarette is filled with a pre-determined amount of e-liquid), the rack assembly **600** is removed from the e-liquid filling apparatus **700**, and the box assemblies **100** removed from the rack assembly **600**. The box assemblies **100** may then be further packaged for shipment directly after being removed from the rack assembly **700** by an operator.

[0070] In use, if an operator adds a certain amount (e.g., five hundred milliliters (500 ml)) of e-liquid (e.g., any liquid solution may contain varying amounts of one or more substances including, but not limited to, nicotine, botanical oil (e.g., hemp oil, cannabis oil (e.g., THC, CBD)), flavorings, and/or other chemicals) into the e-liquid storage container **22**, the operator enters “500” in the “filling amount” on the GUI of the touchscreen display **751**. There are temperature sensors in the storage container, e-liquid pump, and e-liquid filling head **710** to detect the e-liquid temperature of the current position. If the e-liquid is very thick and the fluidity is not good, the operator can pre-heat the e-liquid and then pour the e-liquid into the storage tank. The operator enters the e-liquid temperature into the storage tank in the GUI of the touchscreen display **751**. The amount of e-liquid to be injected in a single time (i.e., into a single electronic cigarette cartridge or atomizer cartridge) needs to be set in “remaining oil” (e.g., 1 milliliter, 1.5 milliliter, etc.), and the controller calculates the volume of each e-liquid injection by controlling the injection pump when executed. The e-liquid filling needle is mounted on the needle valve and the needle holder of the e-liquid filling head **710** must be perpendicular to the rack clamp assembly **730**. The operator adjusts the initial position of the e-liquid filing head **710** via the GUI interface on the display **751** by: (1) pressing the descending arrow on the Z axis of the e-liquid filling assembly to adjust the distance between the needle tip and the atomizer clamp to facilitate the observation of the needle tip position; (2) the e-liquid filling assembly **700** is located to the starting position in the transverse direction (Y-axis) (in the GUI of the display **751**, the operator clicks the forward or backward button several times in the Y-axis until the e-liquid filling needle is aimed at the center of the first hole (located on either the bottom of the rack assembly **600** in the case of an e-liquid filling machine **700** where the electronic cigarette contained within each box assembly is filled from the bottom, or the top of the rack assembly **600**, In the case of an e-liquid filling machine where the electronic cigarette contained within each box assembly is filled from the top), and if this position is exceeded, the forward or

backward button can be clicked several times until it moves to the desired position. And (3) the value is entered in the box after the current “position” of the X axis into the box after the “start position” and press the ENT key in the GUI of the display **751** to confirm. The initial position of the e-liquid filling head **710** is successfully corrected. The distance traveled along the Y-axis after each e-liquid filling and the number of box assemblies **100** can also be set accordingly.

[0071] The e-liquid filling apparatus **700** may be capable of communicating with an external device (not shown) through a communication module contained within the apparatus **700** that provides wired or wireless communication with the external device. In this manner, the apparatus **700** may be remotely controlled (e.g., via an app, website interface, or the like), a user provided with real-time updates, etc. The external device may be a cellphone, a desktop computer, a notebook computer, a personal electronic device, or the like that is located near the apparatus **700** or remote from the apparatus **700**.

[0072] A number of various different types of sensors may be employed by the apparatus **700** before, during, and/or after operation. Various types of temperature sensors may include, without limitation, thermocouples, infra-red (IR) sensors, thermometers, and the like. A temperature sensor may or may not be in contact with the item whose temperature is to be sensed. Various types of sensors for electrical properties may include sensors that can detect or measure impedance, resistance, conductivity, voltage level, current level, and the like. A sensor may be triggered before, during, and/or after operation. according to a detected and/or scheduled event(s). A sensor may be triggered when it receives instructions from a controller. Likewise, any one or more of the sensors may be continuously sensing for the presence of a particular condition(s) and/or may indicate when a particular condition(s) is sensed. A controller may receive signals provided to the controller by one or more sensors where the signals indicate a measured property or properties. The sensor(s) may provide signal(s) to the same controller or to different controllers. The controller may be implemented in some embodiments as a hardware and/or software module which may process sensor signal(s) to interpret the sensor signal(s) for the controller. The signal(s) may be provided to the controller or control assembly **750** via a wired connection, or may be provided wirelessly. The controller or controller assembly **750** may be provided on a system-wide level, group of device level, device level, module level, or component of module level, or any other level as described elsewhere herein. The controller or control assembly **750** may, based on the signals from the sensors, effect a change in a component or maintain the state of a unit. For example, the controller or controller assembly **750** may change the temperature of a thermal control unit, re-position a component, move a box assembly and/or e-liquid filling **710**, and the like. In some embodiments, based on the signal(s) from the sensor(s), the controller or controller assembly **750** may maintain one or more condition of the e-liquid filling device **700**. Signal(s) from the sensor(s) may also permit the controller(s) or control assembly **750** to determine the current state of the e-liquid filling device **700** and track what actions have occurred, or are in progress. This may or may not affect the future actions to be performed by the e-liquid filling device **700**. In some instances, the sensor(s) may be useful for detecting conditions that may include errors or malfunctions of the e-liquid device **700**. The sensor(s) may detect conditions that may lead to an error or malfunction. Sensor(s) may be useful in providing feedback in trying to correct a detected error or malfunction. One or more signal(s) from a single sensor may be considered for particular actions or conditions of the e-liquid filling device. Alternatively, one or more signal(s) from a plurality of sensors may be considered for particular actions or conditions of the e-liquid filling device **700**. The one or more signals may be assessed based on the moment they are provided. Alternatively, the one or more signals may be assessed based on information collected over time. In some embodiments, the controller or control assembly **750** may have a hardware and/or software module which may process one more sensor signals in a mutually-dependent or independent manner to interpret the signals for the controller or control assembly **750**. In some embodiments, multiple types of sensors or detection units may be useful for measuring the same property. In some instances, multiple types

of sensors or detection units may be used for measuring the same property and may provide a way of verifying a measured property or as a coarse first measurement which can then be used to refine the second measurement. The controller may also provide information to an external device. For example, the controller or control assembly **750** may provide a status update to an external device. The controller or control assembly **750** may provide the signals provided by the sensors to the external device. The controller or control assembly **750** may pass on such data as raw data as collected from the sensors. Alternatively, the controller or control assembly **750** may process and/or pre-process the signals from the sensors before providing them to the external device. The controller or control assembly **750** may or may not perform any analysis on the signals received from the sensors. In one example the controller or control assembly **750** may put the signals into a desired format without performing any analysis.

[0073] A Homepage screen may appear on GUI of the display **751** of the e-liquid filling apparatus **700** for controlling and monitoring various aspects of the operation of the e-liquid filling apparatus **700**, and a settings page may appearing on the GUI of the display **751** of the e-liquid filling apparatus **700** for controlling and monitoring various aspects of the operation of the e-liquid filling device **700**, including the positioning, and movement of the e-liquid filling needle of the filling head **710**. The GUI on the display **751** provides touchscreen control. When on the settings page (or any other “page” of the GUI), the user can return to the Homepage screen by touching a “Home” icon at the lower right corner of the GUI of the display **751** and on other screens of the GUI of the display **751**.

[0074] The Homepage and settings page include temperature setting controls for oil, reservoir, pump, pipe, and valve (e.g., the touchscreen capability allows a user to input temperature settings, to adjust temperature settings up or down), and current temperature outputs for current oil temperature, reservoir temperature, pump temperature, pipe temperature, and valve temperature. The Homepage also includes “buttons” for heating on/off, heating lamp on/off, needle position reset, oil filling start, oil filling stop, oil draining/cleaning, exhausting, hint bar (e.g., system failure, etc.), progress tracking, oil (e.g., e-liquid) filling setting, filling motor, guide motor, Z-axis setting, Y-axis setting, Alarm oil, Up delay, Filling enable (debugging mode), Filling value, Reservoir valve, Cleaning, and Exhausting. A “cog” icon or settings icon appearing on the Homepage brings the user to the settings screen or “page” **800** of the GUI when touched by the user. The progress tracking includes Status (e.g., idle, running, working, fault, and other states), Output, and Time. The “idle” state refers to the state where there are no faults and no operations. The “emergency stop” state is where the controller does not issue a command when the emergency stop button is pressed; causing the e-liquid filling device to pause operation. The “operation” state is where the e-liquid filling device normal injects oil and displays information. The “fault” state refers to the state where the controller cannot receive feedback signals from sensor(s) to display. The Output measures filling the e-cigarette or atomizer contained with the box assembly **100** with the preset amount of e-liquid (e.g., botanical oil). The e-liquid storage tank usually stores about two (2) liters of e-liquid, and the total amount of e-liquid filling will be calculated during the e-liquid filling process. When the remaining amount of e-liquid in the storage tank is a certain amount (e.g., about one half (0.5) liter), an alarm indicates that e-liquid needs to be added to the storage tank. With regard to Time setting on the GUI, each e-cigarette or atomizer contained within a box assembly **100** takes about five (5) seconds to be filled with e-liquid. Every time sixteen (16) e-cigarettes or atomizers (each contained within a box assembly (i.e., there are sixteen (16) box assemblies **100** with each one containing an e-cigarette or atomizer)) are to be filled with e-liquid, one by one, about eighty (80) seconds are needed. The oil filling setting includes Current Cart (i.e., indicating which e-cigarette/atomizer is currently being filled (e.g., 1/10, 5 of 16)), Remaining Oil, and Filling Amount. The total number of electronic cigarette cartridges or atomizer cartridges to be filled (i.e., the number of box assemblies **100** (each containing an electronic cigarette or atomizer) to be placed in the rack assembly for filling) is inputted into the e-liquid filling device **700**, before the e-liquid

(e.g., botanical oil) injection operation through the GUI of the touchscreen display **751**. For the purposes of illustration, as a rack assembly **600**, is illustrated that is capable of holding sixteen (16) box assemblies **100**, with a default value of sixteen (16). If necessary, the number of electronic cigarettes or atomizers (each contained within a box assembly **100**) to be filled could be adjusted. That is, when the remaining electronic cigarettes or atomizers to be filled are less than sixteen (16), the rack assembly **600** is installed on the e-liquid filling device **700**, and then the number sixteen (16) is modified to the quantity to be filled through the GUI of the touchscreen display **751**. The quantity of e-liquid is roughly estimated. For example, if 1500 milliliters is initially poured into the storage container, the operator sets a reminder to end the e-liquid injection when there is only 500 milliliters left (at this moment, there may not be only 500 milliliters left in the pump, but the display **751** only indicates that 1000 milliliters of e-liquid has been used). The 1500 milliliters figure is obtained by multiplying the e-liquid injection volume of the valve core each time by the number of e-liquid injection times. At each e-liquid injection, the total amount of 1500 milliliters minus the amount of a single oil injection. The total amount is manually output by the operator, and there is a scale mark inside the pump that can be observed. There may be errors, and e-liquid injection will not be carried out after the e-liquid is completely used, to prevent the occurrence of empty injection.

[0075] When touched by a user, a “WI-FI” icon appearing on the Homepage can bring the user to a screen or “page” (not shown) of the GUI where Link settings, such as wide area network (WAN) settings, WI-FI settings, 4G settings, and basic information regarding connectivity can be found. For example, touching the “Basic Info” “button” on the touchscreen GUI brings the user to the Internet settings page where the GUI can display various information, such as whether the apparatus **700** is online or offline, the internet mode (e.g. Ethernet, 4G, WIFI, etc.), persistent identifier (PID), device serial number, password, firmware, kernal, BOOT, File System, Server, and the like. In another example, touching the “WAN Setting” “button” on the touchscreen GUI brings the user to the WAN setting page where the GUI can display if the WAN interface is configured to obtain an IP address from the Internet Service Provider (ISP) by using Dynamic Host Configuration Protocol (DHCP) or uses a Static Internet Protocol (IP) address, identifies the IP address, primary and secondary DNS servers, Local Area Network (LAN) IP, WAN MAC address, and the like. In another example, touching the Wi-Fi settings “button” on the touchscreen of the GUI brings the user to the Wi-Fi settings page where the GUI can display a service set identifier (SSID), enter a password for the SSID in order to access the SSID, press a WIFI scanning “button” on the touchscreen to identify all WIFI networks that can be detected in the immediate area. In an additional example, touching the 4G settings “button” on the touchscreen of the GUI brings the user to the 4G settings page where the GUI can display the connection between 2G and 4G signals, and remote maintenance can be carried out through WLAN, WIFI, 2G, and 4G signals. For example, for some manufacturers that require customized customization, OTA could be updated remotely. If necessary, machine operating parameters could be remotely monitored. An “IO” icon (representing “Input Output”) appearing on the Homepage can bring the user to an IO Signal screen or “page” of the GUI where Input and Output signals are displayed as either working or not working. The “Input” signals refers to inputting parameters to the controller, and the “Output” signals refer to outputting it to other sensors through the controller. The “Input” signals can include, without limitation, e-liquid injection motor origin, Y-axis motor origin, Z-axis motor origin, start, and emergency stop. The “Output” signals can include, without limitation, Y-axis pulse signal, Z-axis pulse signal, Y-axis movement direction, Z-axis movement direction, pump heating, barrel heating, oil injection valve, pipe heating, and valve heating. The foregoing Input signals and Output signals can be referenced and adjusted. The e-liquid filling device **700** can be controlled through the above input/output IO signals to realize the normal operation of the e-liquid filling machine **700** and ensure the correct and safe operation of the e-liquid filling machine **700**. [0076] In use, a user prepares the e-liquid filling apparatus **700** for use. A user connects a power

cable (not shown) to a power inlet (not shown) located on the housing of the apparatus **700**, and connects an air inlet pipe (not shown) to an air inlet (not shown) on the housing. The user then installs a fresh needle in the e-liquid filling head **710**. The e-liquid storage container is checked to see if any e-liquid remains from previous uses of the apparatus **700** by removing the valve over the e-liquid filling hole. If the storage container contains e-liquid from a prior filling, the storage container is emptied and cleaned. If the storage container is empty and clean, a pre-measured and heated e-liquid formulation is transferred into the storage container, and the valve re-positioned over the e-liquid filling hole. The apparatus **700** is started by pressing a power button (not shown) on the housing. When the apparatus **700** is powered on, the control assembly **750** carries out pre-programmed steps and the GUI appears on the display **751**. The user checks the current temperature and temperature settings on the Homepage screen, and makes any necessary adjustments to the temperature settings. On the Homepage screen, the user touches the “Heater ON/OFF” “button” (which is then reading “Heater OFF” and the button background may be a particular color (e.g., blue)) for heating to start. The button will then say “Heater ON” during the heating process (with the button background being a particular color (e.g., orange) during the heating process) and heating continues until the set temperature is reached, and the heater ON/OFF button reading “Heating ON” (with the button background changing to a different color (e.g., green)) which means heating has finished. The discharge of air from the pipeline between the pump and the e-liquid injection head **710** to prevent inaccurate e-liquid injection is referred to as “exhausting”. It can also be cleaned through the exhaust function by pouring alcohol into the pump. Through the exhaust function, e-liquid in the pipeline is cleaned out between filling different batches of electronic cigarettes or atomizers held within box assemblies **100** to prevent cross odors as different types of e-liquid (having different odors) may be used in different batches. If successive batches of electronic cigarettes or atomizers held within box assemblies **100** are to be filled with e-liquid are to use the same type of e-liquid, then the pipeline does not have to be cleaned between batch fillings.

[0077] Once the exhausting has been completed, and the needle re-installed on the needle head **710**, the needle position is re-set by pressing the “Reset” until “Reset” is replaced by “Ready” on the display **751**, and the apparatus **700** is ready for product loading.

[0078] The rack **700** is then secured in the rack clamp assembly **730** of the apparatus **700**. The user presses a start button on the housing of the apparatus **700** to start the filling process (which had previously been set or programmed using the GUI of the control assembly **750** via the touchscreen display **751**). Prior to touching the start button, the user can check the parameter settings on the display **751** (e.g., to ensure the Filling Amount is set correctly to the tank size of the apparatus **700**).

[0079] Any of the embodiments described with reference to the figures may be implemented, in part, using software, firmware, hardware (e.g., fixed logic circuitry), manual processing, or a combination of these implementations. The terms “logic,” “module,” “component,” “system” and “functionality,” as used herein, generally represent software, firmware, hardware, or a combination of these elements. For instance, in the case of a software implementation, the terms “logic,” “module,” “component,” “system,” and “functionality” represent program code that performs specified tasks when executed on a processing device or devices (e.g., MCU, CPU or CPUs). The program code can be stored in one or more computer readable memory devices. More generally, the illustrated separation of logic, modules, components, systems, and functionality into distinct units may reflect an actual physical grouping and allocation of software, firmware, and/or hardware, or can correspond to a conceptual allocation of different tasks performed by a single software program, firmware program, and/or hardware unit. The illustrated logic, modules, components, systems, and functionality may be located at a single site (e.g., as implemented by a processing device), or may be distributed over a plurality of locations.

[0080] In addition, the claimed invention is not limited in size and may be constructed in various

sizes in which the same or similar principles of operation as described above would apply. Furthermore, the figures (and various components shown therein) of the specification are not to be construed as drawn to scale.

[0081] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. Throughout this specification the word “comprise”, or variations such as “comprises” or “comprising”, will be understood to imply the inclusion of a stated element, integer or step, or group of elements, integers or steps, but not the exclusion of any other element, integer or step, or group of elements, integers or steps. In other words, unless explicitly stated to the contrary, embodiments “comprising” or “having” an element or a plurality of elements having a particular property can include additional elements not having that property. In other words, the terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. In other words, the use of “including,” “comprising,” “having,” “containing,” “involving,” and variations thereof, is meant to encompass the items listed thereafter and additional items. Further, references to “one embodiment” or “one implementation” are not intended to be interpreted as excluding the existence of additional embodiments or implementations that also incorporate the recited features. The term “exemplary” is intended to mean “an example of”.

[0082] As used herein, the singular forms “a”, “an” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. In other words, an element or step recited in the singular and preceded by the word “a” or “an” should be understood as not necessarily excluding the plural of the elements or steps. Further, references to “one embodiment” or “one implementation” are not intended to be interpreted as excluding the existence of additional embodiments or implementations that also incorporate the recited features. Thus, when introducing elements of aspects of the disclosure or the examples thereof, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. In other words, the indefinite articles “a”, “an”, “the”, and “said” as used in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The use of the expression “at least” or “at least one” suggests the use of one or more elements or ingredients or quantities, as the use may be in the embodiment of the disclosure to achieve one or more of the desired objects or results.

[0083] The numerical values mentioned for the various physical parameters, dimensions or quantities are only approximations and it is envisaged that the values higher/lower than the numerical values assigned to the parameters, dimensions or quantities fall within the scope of the disclosure, unless there is a statement in the specification specific to the contrary. Any range or value given herein can be extended or altered without losing the effect sought, as will be apparent to the skilled person.

[0084] When an element or layer is referred to as being “on”, “engaged to”, “connected to” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to”, “directly connected to” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0085] In the description of the present invention, several means one or more, a plurality of means more than two, greater than, less than, more than, and the like are understood as not including this number, while above, below, within, and the like are understood as including this number. If there

are the descriptions of first and second, it is only for the purpose of distinguishing technical features, and should not be understood as indicating or implying relative importance, implicitly indicating the number of the indicated technical features or implicitly indicating the order of the indicated technical features.

[0086] In the description of the present invention, it should be noted that the terms “installation”, “connected” and “connection” if any shall be understood in a broad sense unless otherwise specified and defined. For example, they may be fixed connection, removable connection or integrated connection; may be mechanical connection or electrical connection; and may be direct connection, or indirect connection through an intermediate medium, and connection inside two elements. The specific meanings of the above terms in the present invention can be understood in a specific case by those of ordinary skills in the art.

[0087] While various spatial and directional terms, such as “up”, “down”, “front”, “rear”, “left”, “right”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, “upper,” “lower,” and the like are used to describe embodiments and implementations of the present disclosure, it is understood that such terms are merely used with respect to the orientations shown in the drawings. The orientations can be inverted, rotated, or otherwise changed, such that a top side becomes a bottom side if the structure is flipped 180 degrees, becomes a left side or a right side if the structure is pivoted 90°, and the like. In other words, spatially relative terms, such as “inner,” “outer,” “beneath”, “below”, “above”, “lateral”, “longitudinal” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

[0088] In the description of the present invention, it should be understood that the orientation or position relationship indicated by the terms is based on the orientation or position relationship shown in the accompanying drawings, it is only for the convenience of description of the present invention and simplification of the description, and it is not to indicate or imply that the indicated device or element must have a specific orientation, and be constructed and operated in a specific orientation. Therefore, the terms shall not be understood as limiting the present invention.

[0089] As used herein, a structure, limitation, or element that is “configured to” perform a task or operation is particularly structurally formed, constructed, or adapted in a manner corresponding to the task or operation. For purposes of clarity and the avoidance of doubt, an object that is merely capable of being modified to perform the task or operation is not “configured to” perform the task or operation as used herein.

[0090] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

[0091] It will be understood that the benefits and advantages described above can relate to one embodiment or can relate to several embodiments. The embodiments are not limited to those that solve any or all of the stated problems or those that have any or all of the stated benefits and advantages. It will further be understood that reference to ‘an’ item refers to one or more of those items.

[0092] The order of execution or performance of the operations in examples of the disclosure illustrated and described herein is not essential, unless otherwise specified. That is, the operations can be performed in any order, unless otherwise specified, and examples of the disclosure can

include additional or fewer operations than those disclosed herein. For example, it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation (e.g., different steps, etc.) is within the scope of aspects and implementations of the disclosure. In other words, the method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0093] The phrase “one or more of the following: A, B, and C” means “at least one of A and/or at least one of B and/or at least one of C.” The phrase “and/or”, as used in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0094] As used in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of” “only one of” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0095] As briefly discussed above, as used in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0096] Use of ordinal terms such as “first,” “second,” “third,” etc., in the claims to modify a claim element does not by itself connote any priority, precedence, or order of one claim element over another or the temporal order in which acts of a method are performed. Ordinal terms are used merely as labels to distinguish one claim element having a certain name from another element having a same name (but for use of the ordinal term), to distinguish the claim elements.

[0097] Having described aspects of the disclosure in detail, it will be apparent that modifications and variations are possible without departing from the scope of aspects of the disclosure as defined

in the appended claims. As various changes could be made in the above constructions, products, and methods without departing from the scope of aspects of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawings shall be interpreted as illustrative and not in a limiting sense.

[0098] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) can be used in combination with each other. In addition, many modifications can be made to adapt a particular situation or material to the teachings of the various embodiments of the disclosure without departing from their scope. While the dimensions and types of materials described herein are intended to define the parameters of the various embodiments of the disclosure, the embodiments are by no means limiting and are example embodiments. Many other embodiments will be apparent to those of ordinary skill in the art upon reviewing the above description. The scope of the various embodiments of the disclosure should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112 (f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

[0099] This written description uses examples to disclose the various embodiments of the disclosure, including the best mode, and also to enable any person of ordinary skill in the art to practice the various embodiments of the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the various embodiments of the disclosure is defined by the claims, and can include other examples that occur to those persons of ordinary skill in the art. Such other examples are intended to be within the scope of the claims if the examples have structural elements that do not differ from the literal language of the claims, or if the examples include equivalent structural elements with insubstantial differences from the literal language of the claims.

[0100] The above description presents the best mode contemplated for carrying out the present invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains to make and use this invention. This invention is, however, susceptible to modifications and alternate constructions from that discussed above that are fully equivalent. Moreover, features described in connection with one embodiment of the invention may be used in conjunction with other embodiments, even if not explicitly stated above. Consequently, this invention is not limited to the particular embodiments disclosed. On the contrary, this invention covers all modifications and alternate constructions coming within the spirit and scope of the invention as generally expressed by the following claims, which particularly point out and distinctly claim the subject matter of the invention.

[0101] The following claims are thus to be understood to include what is specifically illustrated and described above, what is conceptually equivalent, what can be obviously substituted and also what essentially incorporates the essential idea of the invention. Those skilled in the art will appreciate that various adaptations and modifications of the just-described preferred embodiment can be configured without departing from the scope of the invention. The illustrated embodiment has been set forth only for the purposes of example and that should not be taken as limiting the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

[0102] Various technical features of the above embodiments may be combined randomly, and in order to simplify the description, possible combinations of various technical features in the above

embodiments are not all described. However, as long as the combinations of these technical features have no contradiction, the combinations of these technical features should be considered as falling into the scope recorded by the specification.

[0103] Although the embodiments of the present invention have been shown and described, those of ordinary skills in the art may understand that various changes, modifications, substitutions and variations may be made to these embodiments without departing from the principle and purpose of the present invention, and the scope of the present invention is defined by the claims and their equivalents.

[0104] The following claims are thus to be understood to include what is specifically illustrated and described above, what is conceptually equivalent, what can be obviously substituted and also what incorporates the essential idea of the invention. Those skilled in the art will appreciate that various adaptations and modifications of the just-described preferred embodiment can be configured without departing from the scope of the invention. The illustrated embodiment has been set forth only for the purposes of example and that should not be taken as limiting the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

Claims

1. A box assembly, comprising: a body portion; a tray portion disposed within the body portion; and a cover portion disposed over the tray portion and body portion; wherein the body portion includes a first filling port at one end, and the tray portion includes a second filling port aligned with the first filling port.
2. The box assembly of claim 1, wherein the second filling port tray includes an open end of the tray portion aligned with the first filling port.
3. The box assembly of claim 1, wherein the body portion includes an accommodating cavity sized and shaped to match a size and shape of the tray portion.
4. The box assembly of claim 1, wherein the cover portion is configured to removably engage the body portion.
5. The box assembly of claim 1, wherein the tray portion comprises an interior portion sized and shaped to receive a device; and wherein the device includes a filling port aligned with the second filling port of the tray portion when disposed within the tray portion.
6. The box assembly of claim 5, wherein the filling port of the device is aligned with the first filling port of the body portion
7. The box assembly of claim 1, wherein the cover portion includes a plurality of recessed perimeter walls set back on all sides from respective side edges of the cover portion; wherein the perimeter walls extend downward from a bottom surface of the cover portion and are positioned to be received within a recessed portion of the body portion such that a portion of a bottom surface of the cover portion abuts against a top surface of the body portion.
8. The box assembly of claim 1, wherein the cover includes a plurality of bulges configured to press-fit engage an interior portion of the body portion.
9. A rack assembly configured for holding at least one box assembly device in position for providing a pre-determined amount of e-liquid to a device having a filling port contained within the at least one box assembly, comprising: an upper cover; a lower cover; and a limit member disposed between the upper and lower covers; wherein the upper cover, lower cover, and limit member are configured to vertically align the at least one box assembly; wherein at least one of the upper and lower covers includes at least one filling hole; wherein the at least one box assembly includes a body portion, a tray portion disposed within the body portion, and a cover portion disposed over the tray portion and body portion; and wherein the body portion includes a first filling port at one end, the tray portion includes a second filling port, and the first and second filling ports are aligned

with the at least one filling hole and the filling port of the device.

10. The rack assembly of claim 9, wherein the rack assembly further comprises first and second support members; wherein the first and second support members engage the upper and lower covers.

11. The rack assembly of claim 9 wherein the upper cover includes at least one upper limit groove, with the at least one upper limit groove configured to receive and engage a portion of the at least one box assembly at a first end of the at least one box assembly.

12. The rack assembly of claim 9, wherein the lower cover includes at least one lower limit groove, with the at least one lower limit groove configured to receive and engage a portion of the at least one box assembly at a second end of the at least one box assembly

13. A method for providing e-liquid to at least one device having a filling port contained within a box assembly, wherein the at least one device includes a reservoir in communication with the filling port, and wherein the at least one box assembly includes a body portion, a tray portion disposed within the body portion, and a cover portion disposed over the tray portion and body portion, wherein the body portion includes a first filling port at one end, the tray portion includes a second filling port, and the first and second filling ports are aligned with the at least one filling hole and the filling port of the device, comprising: providing an e-liquid filling apparatus comprising a workbench; an e-liquid filling assembly configured to provide e-liquid to the reservoir of the at least one device contained within the box assembly; a moving assembly configured to adjust a filling portion of the e-liquid filling assembly and the box assembly relative to each other; and a control assembly configured to control operation of the e-liquid filling apparatus; configuring the control assembly to fill the at least one device contained within the box assembly with e-liquid, including setting a pre-determined amount of e-liquid the e-liquid filling assembly provides to the at least one device; positioning the at least one device in a rack assembly configured to hold the at least one device; operationally engaging the rack assembly with the workbench; and filling the at least one device contained within the box assembly with the pre-determined amount of e-liquid.
